

KARTIKEYA VEMULA

Data Analyst | SQL | Power BI | Python | dbt | Analytics Engineering

vnskartikeya@gmail.com | +1 (443) 630-1157

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Open to relocate across India

Data Analyst with strong **SQL**, **Power BI**, **Python**, and **dbt** experience building warehouse-backed KPI marts, diagnostic workflows, and decision-support reporting layers. GCP-certified in data engineering, with hands-on experience translating raw datasets into business-ready analytics, root-cause insights, and stakeholder recommendations.

EXPERIENCE

Graduate Research Assistant, Risk Analytics — UMBC, Baltimore

Oct 2025 – Jan 2026

Project: Extreme Value AQI Risk Modeling, Baltimore

- Built an EPA AirData **ETL pipeline** that converted raw air-quality data into a validated dataset of **6,573 records** spanning 2006 to 2023 across AQI, pollutant, and meteorological variables.
- Designed structured **data quality checks** across missing values, outliers, and station-level reporting inconsistencies to keep the dataset defensible for risk modeling and stakeholder review.
- Modeled pollutant risk using **XGBoost**, achieving **RMSE 8.99** and **R² 0.852**, giving a reliable baseline for expected AQI behavior across historical observations.
- Applied Extreme Value Theory using a Generalized Pareto Distribution across 98 AQI exceedance events, quantifying a **6.65% conditional tail-risk probability** for extreme pollution episodes.
- Translated model outputs into **stakeholder-ready risk reporting**, framing exceedance likelihood and tail-risk findings for non-technical monitoring and resource-prioritization discussions.

SKILLS

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|------------|-----------------------|-------------------------|
| • SQL | • PostgreSQL | • KPI Reporting |
| • Power BI | • BigQuery | • Data Quality Testing |
| • Python | • dbt Core | • Root Cause Analysis |
| • DAX | • Data Warehousing | • Stakeholder Reporting |
| • Excel | • ETL / ELT Pipelines | • Power Query |

CERTIFICATIONS

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| • Microsoft PL-300: Power BI Data Analyst Associate | • Snowflake Data Warehousing Workshop |
| • Google Cloud Certified: Professional Data Engineer | • Google Analytics Certification |
| • Google Cloud Certified: Associate Cloud Engineer | • Cisco Data Analytics Essentials |
| • dbt Fundamentals | • Kaggle Advanced SQL |

PROJECTS

Customer Intelligence Data Warehouse

Python, PostgreSQL, Docker, dbt Core, SQL, Star Schema, Power BI, DAX, Power Query

- Converted noisy retail transactions into tested **dbt marts** for executive KPIs, customer lifetime value, repeat purchase, country revenue, and monthly sales, giving stakeholders governed definitions for **£3,279,364.61** in revenue across **5,047** known customers.
- Modeled repeat-purchase and customer-value intelligence across the customer base, identifying **3,394 repeat buyers**, a **67.25% repeat purchase rate**, and a top customer worth **£62,131.42** to support retention prioritization.

- Built country revenue marts showing UK concentration at **£2,782,932.74** and **84.86%** of total revenue, enabling geographic performance monitoring beyond aggregate sales reporting.
- Delivered a **published interactive Power BI dashboard** with BI-ready reporting exports, connecting KPIs, repeat purchase behavior, customer value, and country revenue into a single decision-support layer.

KPI Reliability and Diagnostic Engine

Python, PostgreSQL, dbt Core, SQL, Streamlit, Anomaly Detection, Root Cause Analysis

- Built a KPI diagnostic engine monitoring **121,500 metric rows** across **5 KPIs**, converting raw metric movement into validated anomaly, **RCA**, and incident outputs for stakeholder triage.
- Generated **12 incident records** with severity, confidence, root-cause segment, and recommended investigation paths, helping prioritize which KPI movements deserved immediate review.
- Diagnosed a **CRITICAL latency incident** where latency rose from an expected **158.9 ms** to an actual **343.3 ms**, with RCA isolating North America, Desktop, and Chrome as **100.0%** of excess latency.
- Compared threshold, rolling Z-score, residual, and combined detectors, selecting residual detection after improving **F1 to 0.5485** and reducing false positives from **2,958 to 563** on synthetic evaluation labels.

Instacart Intelligence Platform

Python, PostgreSQL, Docker, dbt Core, SQL, Power BI, Cohort Analysis, Basket Analysis

- Engineered PostgreSQL and **dbt marts** over **3,421,083 orders** and **33,819,106 order-item rows** to model customer lifecycle, product performance, retention milestones, and basket affinity for retail decision support.
- Built behavioral customer segmentation identifying **69,635 Moderate Engagement**, **62,818 Loyal Routine**, and **44,873 At Risk** customers, enabling lifecycle targeting and retention prioritization.
- Modeled order-sequence retention showing customer continuation dropped to **53.70%** by **order 10**, giving stakeholders a concrete milestone for reorder nudges and lifecycle campaigns.
- Packaged findings into a **published interactive Power BI dashboard** and recommendation memo, including basket affinity analysis that separated broad cross-sell pairs from niche high-lift recommendation opportunities.

Growth Funnel Intelligence Engine

Python, SQL, PostgreSQL, Plotly Dash, Funnel Analysis, Product Analytics, A/B Testing

- Modeled GA4 ecommerce events into session-level funnel stages, identifying a **75.41%** Session Start to Product View drop-off and prioritizing product discovery as the highest-leverage optimization area.
- Built **SQL-based** funnel, attribution, and cohort views across acquisition channel, device, and engagement depth, separating conversion quality from raw conversion volume.
- Analyzed acquisition and engagement patterns, finding direct traffic as the highest-converting channel and High Intent 20+ minute sessions as a safer **25.21% conversion** cohort across **932 sessions**.
- Evaluated session-level A/B test results with **two-proportion z-tests**, finding Control **1.75%** vs Variant **1.65%** macro CVR, **-5.49% lift**, **p=0.48154**, and recommending no rollout.

EDUCATION

M.S. Data Science — University of Maryland Baltimore County, USA

2024 – 2026

GPA: **3.78** out of 4.0 | Focus: Machine Learning, Statistical Modeling, Experimental Design, Time Series Analysis

B.Tech Computer Science (AI & ML) — GITAM University, India

2020 – 2024

GPA: **8.89** out of 10, First Class with Distinction

EXTRA-CURRICULAR AND LEADERSHIP

- Mentored peers across Mathematics, Digital Logic, Data Communication, Design Thinking, and Introduction to Data Analysis and Machine Learning, supporting analytical reasoning and technical concept clarity.
- Led a 3-part UMBC peer-teaching presentation series on machine learning concepts, covering terminology, model training, technical explanation, and real-world case studies.
- Supported chess club event coordination and drafted social media content as part of the student organizing team, contributing to event visibility and student engagement.